

Zoey Situ

[LinkedIn](#) | zoeysitu05@gmail.com | 778-706-2005

TECHNICAL SKILLS

Programming: C++, Python, MATLAB, VHDL, RISC-V assembly

Embedded & Hardware: STM32, RP2040, UART/I2C/SPI/ARINC-429, PCB Design (KiCad, Altium), FPGA Development

Industrial & Systems: Allen-Bradley PLCs, HMIs, Industrial I/O, Ethernet/IP, VLANs, BIM 360, Bluebeam Revu

Tools & Test Equipment: Oscilloscope, DMM, Logic Analyzer, Soldering/SMD Rework, LabVIEW, Function Generator, LTspice

EDUCATION

Bachelor of Applied Science in Electrical Engineering

Sept. 2023 – Present

University of Waterloo, Waterloo, ON

- **Relevant Courses:** RF & Microwave Circuits, Computer Networks, Electronic Devices, Electronic Circuits, Communication Systems, Electromagnetic Fields & Waves, Analog Control Systems, Semiconductor Devices

PROFESSIONAL EXPERIENCE

Building Systems Student

Jan. 2026 – May 2026

PCL Construction Westcoast Inc.

Vancouver, BC

- Coordinated field verification and readiness tracking for **Electrical, ESS, IMIT**, and low-voltage infrastructure across telecom rooms, electrical rooms, and healthcare facility spaces using **BIM 360, CUIX**, Bluebeam Revu, and DroneDeploy.
- Performed technical inspections and site validation for **access control**, nurse call, **BMS interfaces**, **cable pathways**, **HVAC integration**, and environmental compliance systems within active hospital construction environments.
- Monitored telecom room conditions and temporary infrastructure systems, including cooling units, air scrubbers, **humidity/temperature sensors**, and IPAC compliance using **Job Site Insights (JSI)** and field instrumentation workflows.
- Managed **RFIs, submittals**, coordination markups, deficiency tracking, and multidisciplinary issue resolution using **PM4+**, BIM 360, Excel, and Bluebeam to support mechanical, electrical, and IMIT system integration workflows.
- Supported owner walkthroughs, **BIM coordination reviews**, and subcontractor inspections by documenting field conditions, validating installation progress, and resolving construction and system coordination deficiencies.

Hardware and Embedded Systems Research Co-op

May 2025 – Aug. 2025

Safety Critical Real-Time Embedded System Laboratory

Waterloo, ON

- Designed and optimized a **current-sensing PCB** in KiCad for low-voltage CPU power measurement, improving signal stability and magnetic sensitivity through trace optimization, grounding strategies, and **EMI mitigation techniques**.
- Integrated **USB-C, Micro-USB, GPIO**, and sensor interfaces while developing **LabVIEW**-based validation workflows and mechanical interface components in **Fusion 360**.
- Reengineered a treadmill-based RC vehicle control board by replacing an Arduino platform with a **Raspberry Pi Pico W (RP2040)**, reducing PCB size and improving wireless communication reliability via **UART** and **Wi-Fi/Bluetooth**.
- Designed a **4-layer embedded control PCB** integrating **MOSFET drivers, buck converters, and level shifters** to support real-time motor control and obstacle avoidance system integration.

Engineering Co-op

Sept. 2024 – Dec. 2024

Bunge North America

Hamilton, ON

- Supported multiple automation and maintenance projects within active industrial production environments, including assisting with the integration of a solenoid valve actuator into **Allen-Bradley PLC** and **HMI-controlled systems**.
- Assisted with OT and plant data organization by developing **Excel** and **VBA**-based workflows to structure equipment tagging, automation records, and process data for **Google Cloud AI** integration and analytics applications.
- Worked with industrial automation infrastructure including **Ethernet/IP** communication, **industrial I/O systems**, VLANs, network switches, routers, and **function block diagram (FBD)** workflows supporting plant operations and control systems.

PROJECTS

ARINC-429 Trigger & Interface Board

Safety Critical Real-Time Embedded System Laboratory

May 2026 – Aug 2026

- Designed and assembled a **4-layer ARINC-429 interface PCB** in KiCad for avionics fuzzing/testing, enabling passive bus monitoring and message-based triggering.
- Integrated the **HI-35850 ARINC-429 transceiver** with an **STM32** via **SPI** to decode bus messages and support configurable data analysis and trigger generation.
- Designed **3.3 V/5 V power distribution, analog/digital grounding**, decoupling, and trigger interfaces for reliable mixed-signal operation.
- Completed schematic capture, **custom symbols/footprints**, **PCB layout**, **ERC/DRC validation**, fabrication, and **SMD assembly** for system bring-up.

PKD-Agnostic Subthreshold CMOS Voltage Reference

May 2026 – Present

UWASIC Design Team

- Co-designed a **subthreshold CMOS voltage reference**, integrating **PTAT/CTAT compensation**, a **two-stage error amplifier**, and startup circuitry using the **SKY130 PDK**.
- Designed and characterized the error amplifier, then collaborated on full-system **integration, biasing, transistor sizing, and debugging in Xschem/ngspice**.
- Achieved **~0.63 V output, 89 ppm/°C** across **-40°C to 120°C**, and **~65 dB low-frequency PSRR** through temperature and frequency-domain simulations.
- Collaborated on comparing the transistor-level reference against a **Verilog-A behavioural model** to evaluate temperature and PSRR trends.

Embedded Systems Control Project

Oct. 2025

University of Waterloo

Waterloo, ON

- Programmed and configured an **STM32 Nucleo F401RE** MCU using **STM32CubeIDE** to implement timers, **PWM, ADC, UART, and interrupt-driven peripheral control** for real-time embedded sensing and motor control applications.
- Integrated embedded hardware modules including **DC/servo motors, ultrasonic sensors, RPM sensing, RGB LEDs**, and **5V ↔ 3.3V level-shifted** communication interfaces with synchronized peripheral interaction.
- Debugged and validated system operation through **oscilloscope waveform analysis and DMM testing** to verify PWM timing, signal integrity, and voltage translation across hardware interfaces.